IEEE



Texas State IEEE Student Branch

OPS project #5

Today's project is Pitch Perfect

demo



Session Goals

By the end of today, students will:

- Understand the difference between digital and analog inputs
- Learn how a potentiometer works as a voltage divider
- Use analogRead() and map() to convert a sensor reading into a useful range
- Use tone() and noTone() to drive a piezo buzzer at different frequencies
- Build a working pitch controller that plays a note only while the button is held

New part: Potentiometer

What is it?

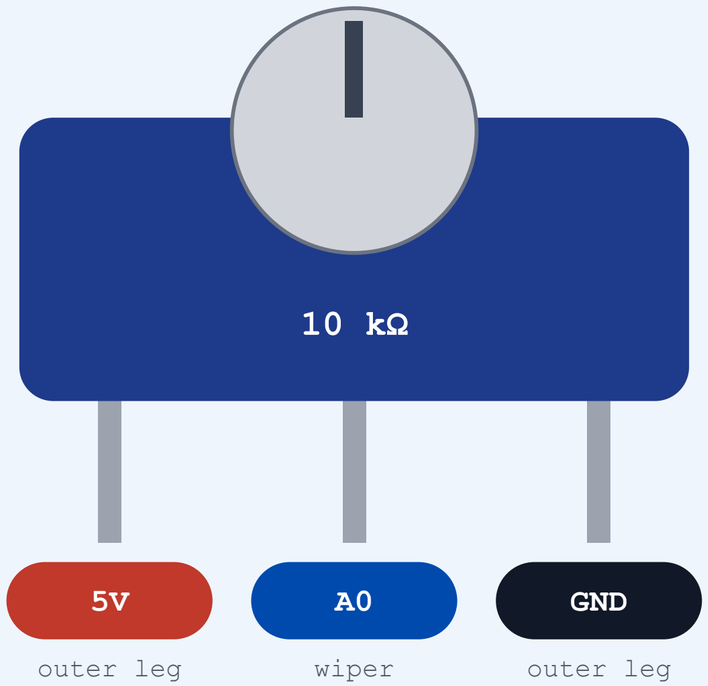
- A knob that slides a contact along a resistor.
- A voltage divider: 0 V one end, 5 V the other.

How to wire it:

- Outler legs -> 5V and GND
- Middle leg -> A# (Any Analog GPIO)

How to wire it:

- analogRead(A#); -> 0 to 1023
- map(); -> turns the Read into a frequency



Turn the knob → the middle pin moves from 0 V to 5 V → analogRead gives 0–1023

Good Terms to Know

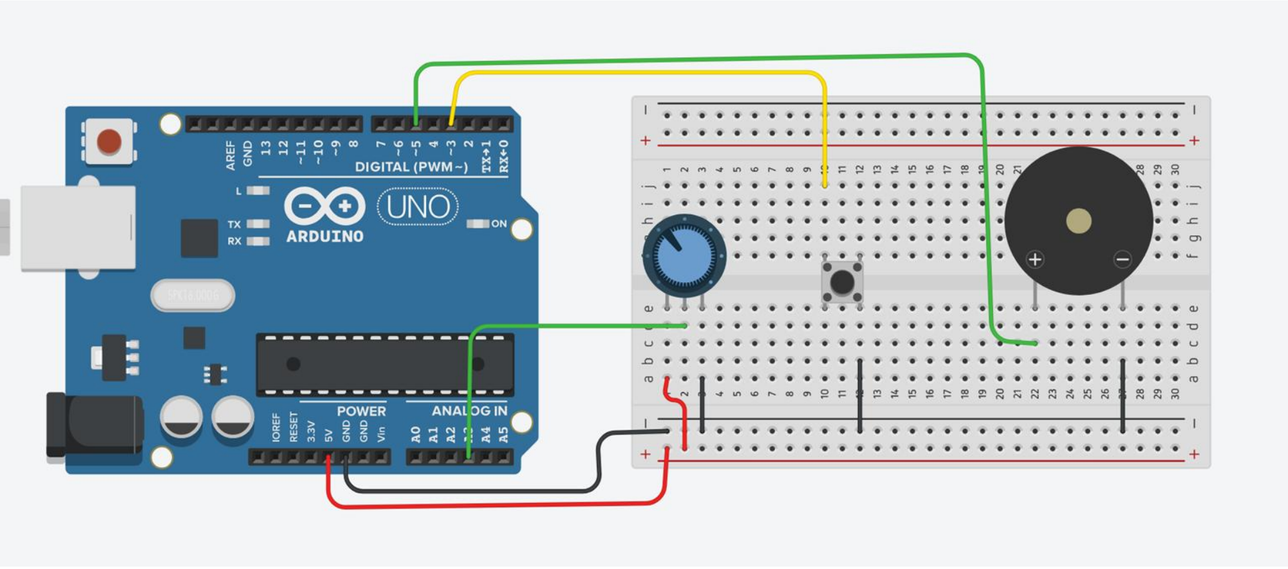
Mapping (map())

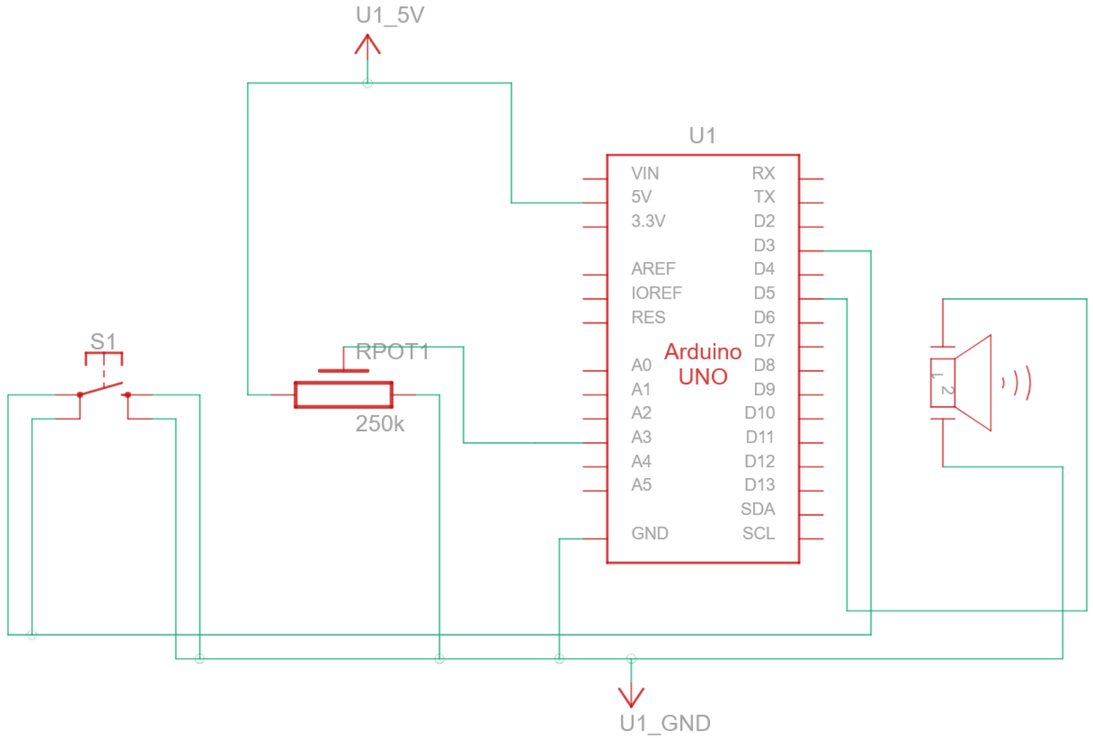
- A function that converts a number from one range into another range.
- The knob gives 0 to 1023, but the buzzer needs a pitch in hertz (Hz), like 100 to 2000.
- map(potValue, 0, 1023, 100, 2000) stretches one range to fit the other, so halfway on the knob is about halfway up the pitch range.

Analog Input

- A signal that can be any value in a range, not just on or off like a button.
- analogRead() measures the voltage on an analog pin (0 to 5 V) and turns it into a number from 0 to 1023.
- Turning the potentiometer changes that voltage, which is how the knob controls the pitch.

Circuit diagrams





Pseudocode

```
// Create three constants:

// Choose a frequency range

void setup() {
  // Set up the pins and Serial

}


void loop() {
  // Read the knob and convert it to a frequency
  // use analogRead()
  // use map()

  // Play or stop based on the button
  // - if the button reads LOW make sound, otherwise don't.



  // Print the values
  // - print the pot value and the frequency to the Serial
Monitor
  // - add a short delay

}
```